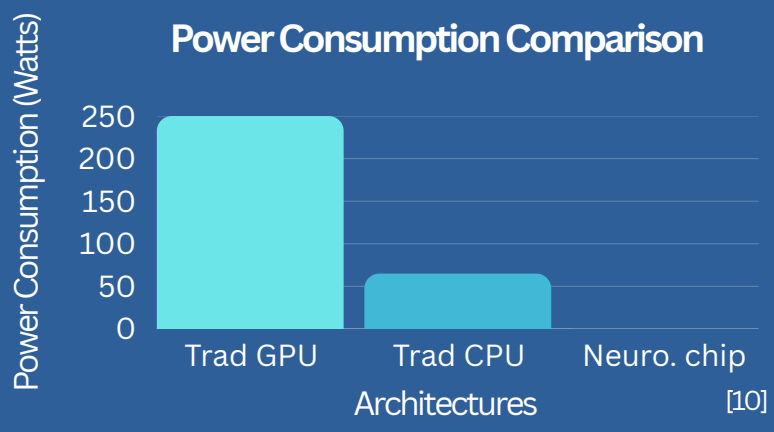
Neuromorphic Computing in Edge A.I.

The Problem:

Regular computer chips use too much power to run AI on small devices

The Solution:

Utilize neuromorphic computing technology to enable energy-efficient edge AI directly on battery-powered devices



[10]

Hardware

Memristors

- Store and compute data simultaneously
- Variable resistance mimics synaptic weights

[8]

Crossbar Arrays

Arrange memristors in grids for massive parallel processing (architecture performs in-memory computing).

[8]

Software

Surrogate Gradient

- non-differentiable → differentiable one
- More efficient by Binarized Neural Networks

STDP: Spike Timing Dependent Plasticity

- Synaptic strength → relative timing of spikes between neurons
- Energy-efficient

[5],[6]

SNN (Spiking Neural Networks)

- Spikes transmit and process information
- Extremely energy-efficient

[3]

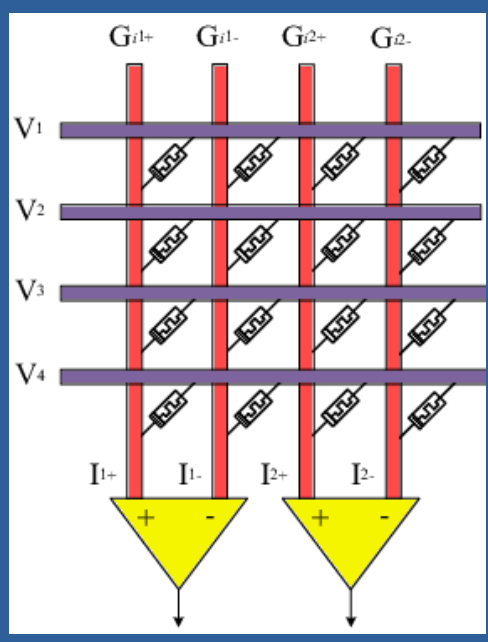


Fig. 1. Neural network implementation using memristor crossbar arrays [9].

Applications

Healthcare

Industrial

Aerospace

Agriculture

Logistics [1]

Results (NeuRRam chip)

40-60% lower latency [2]

70% reduction in image reconstruction [11]

1.6 - 2.3 x higher efficiency [11]

Challenges

- Fabrication and scalability challenges
- Standardization issues
- Ethical aspects considerations

Ethical Aspects

Autonomous decision-making

Job displacement

1000× efficiency & Lower emissions
Cleaner, faster, and capable of running AI anywhere [2]

BIBLIOGRAPHY

- [1]D. Hoffman, "Neuromorphic computing and the future of edge AI," CIO, Sep. 05, 2025. https://www.cio.com/article/4052223/neuromorphic-computing-and-the-future-of-edge-ai.html
- [2]"Neuromorphic vs Traditional Chips: Signal Processing Speed," Patsnp Eureka, 2022. https://eureka.patsnap.com/report-neuromorphic-vs-traditional-chips-signal-processing-speed (accessed Oct. 31, 2025).
- [3] H. Zheng and Y. Yi, "Spiking neural encoding schemes and STDP training algorithms for edge computing," in Proceedings of the Eighth ACM/IEEE Symposium on Edge Computing (SEC 2023), pp. 365-371, 2023, doi: 10.1145/3583740.3626816
- [4] Y. Lu and A. Sengupta, "Deep unsupervised learning using spike-timing-dependent plasticity," Neuromorphic Computing and Engineering, vol. 4, no. 1, 2024, doi: 10.1088/2634-4386/ad3a95.
- [5]B. Cramer et al., "Surrogate gradients for analog neuromorphic computing," Proceedings of the National Academy of Sciences, vol. 119, no. 4, Jan. 2022, doi: 10.1073/pnas.2109194119.
- [6]D. Devansh, "Using Surrogate Gradients and STE in Machine Learning," Machine-Learning-Made-Simple (Medium), Aug. 17, 2024. [Online]. Available: https://machine-learning-made-simple.medium.com/using-surrogate-gradients-and-ste-in-machine-learning-0f949d01dd78. [Accessed: 24-Nov-2025]
- [7] "Hardware implementation of the proposed two-layers memristor crossbar... | download scientific diagram," Research Gate, https://www.researchgate.net/figure/Hardware-implementation-of-the-proposed-two-layers-memristor-crossbar-array-A-3002105_fig2_328984146 (accessed Nov. 24, 2025).
- [8] B. Jin, Z. Wang, T. Wang, and J. Meng, "Memristor-Based Artificial Neural Networks for Hardware Neuromorphic Computing," Natl. Libr. Med. (Natl. Cent. Biotechnol. Inst.), Jul. 2025. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC12231232/ [Accessed: Oct. 19, 2025].
- [9] K. Sayed, "Neural network implementation using memristor crossbar arrays," ResearchGate. [Online]. Available: https://www.researchgate.net/figure/Neural-network-implementation-using-memristor-crossbar-arrays-a-shows-a-conventional_fig1_328984146. [Accessed: Nov. 20, 2025].
- [10] "A power analysis of neuromorphic chips versus GPUs for AI tasks," PatSnap Eureka. [Online]. Available: https://eureka.patsnap.com/report-a-power-analysis-of-neuromorphic-chips-versus-gpus-for-ai-tasks. [Accessed: Nov. 20, 2025].
- [11] University of California – San Diego, "A new neuromorphic chip for AI on the edge, at a small fraction of the energy and size of today's compute platforms," ScienceDaily, 26 Aug. 2022. [Online]. Available: https://www.sciencedaily.com/releases/2022/08/220817114253.htm. Accessed: 24 Nov. 2025